# Primitive Root

## Definition

In modular arithmetic, a number $g$ is called a `primitive root modulo n` if every number coprime to $n$ is congruent to a power of $g$ modulo $n$. Mathematically, $g$ is a `primitive root modulo n` if and only if for any integer $a$ such that $\gcd(a, n) = 1$, there exists an integer $k$ such that:

$$g^k \equiv a \pmod n.$$

$k$ is then called the `index` or `discrete logarithm` of $a$ to the base $g$ modulo $n$. $g$ is also called the `generator` of the multiplicative group of integers modulo $n$.

In particular, for the case where $n$ is a prime, the powers of primitive root runs through all numbers from $1$ to $n - 1$.

## Existence

Primitive root modulo $n$ exists if and only if:

- $n$ is 1, 2, 4, or
- $n$ is power of an odd prime number $(n = p^k)$, or
- $n$ is twice power of an odd prime number $(n = 2 \cdot p^k)$.

This theorem was proved by Gauss in 1801.